

Validity Analysis: MILU

Target context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation
Deployment
Overall risk: **MEDIUM**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking teacher in India uses an LLM to evaluate student responses in an examination setting, assign scores with the support of an AI system, and provide feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation at a subsequent stage.

Target population: A postgraduate- or PhD-qualified teacher with 0–20+ years of teaching experience in Hindi, based in North India and interacting in the Hindi language.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	3	Moderate gaps	medium
Input Content $\triangle$	2	Significant gaps	high
Input Form $\checkmark$	4	Minor gaps	high
Output Ontology $\checkmark$	4	Minor gaps	high
Output Content $\triangle$	2	Significant gaps	medium
Output Form $\checkmark$	5	Strong alignment	high
Average	3.3		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU presents a structurally compatible MCQ benchmark for the North Indian Hindi-medium teacher deployment — the format, output schema, scoring metric, and Devanagari script all match the deployment exactly, and Hindi is the highest-resource Indic

language in MILU's scope. However, the HIGH-priority Input Content dimension reveals a critical empirical concern: 100% of sampled Hindi validation items are machine-translated from English (is_translated=True), far exceeding the dataset-wide 25% translation figure cited in the paper. Translated items use English transliterations for technical vocabulary that may not align with state-board Hindi conventions, the only sampled Literature item is about Wordsworth (not canonical Hindi authors), and only one item explicitly references North Indian state content. Combined with absent annotator demographics and no published per-state Hindi-source distribution, the benchmark provides weaker evidence than its India-first framing suggests for the specific construct of evaluating Hindi-medium board-aligned MCQ comprehension. Output ontology, output form, and input form are well-aligned; input ontology is partially aligned with notable gaps in school-literature framing and state civics; input content and output content are the dimensions where validity is most attenuated.

Practical Guidance

What This Benchmark Measures

MILU validly measures an LLM's ability to answer single-correct-answer Hindi MCQs sourced from (or translated to look like) Indian competitive-exam content across 8 broad domains, with reliable binary-scored output that matches the deployment's positive/negative marking schema. The strongest dimensions — Output Form (5), Output Ontology (4), and Input Form (4) — confirm that the benchmark's scoring infrastructure and Hindi/Devanagari modality are a direct fit for this deployment.

Construct Depth

Construct depth is shallow-to-moderate for the deployment's actual target construct (Hindi-medium board-aligned MCQ evaluation). The benchmark probes pan-India competitive-exam-style knowledge in formal Khari Boli, but the dataset analysis reveals that the Hindi validation split is dominated by GPT-4O translations from English source items, which means MILU primarily tests translated-English knowledge rendered in Hindi rather than natively-sourced Hindi-medium curricular knowledge. Coverage of canonical Hindi-board literature (Tulsidas, Premchand), state-specific civics (UP/MP/Rajasthan/Bihar Panchayati Raj), and school-register content (vs. competitive-GK register) is empirically weak in the sampled data.

What Else You Need

To complete the assessment for this deployment, supplement MILU with: (1) a curated Hindi-board test set using items from UP/MP/Rajasthan/Bihar/CBSE Hindi-medium board exams (addresses Input Content and Input Ontology gaps); (2) a Hindi literature MCQ set covering canonical authors in school-literature register (addresses the Wordsworth-vs-Tulsidas Arts & Humanities gap); (3) a state civics and Panchayati Raj MCQ set (addresses Input Ontology under-representation); (4) Hindi-medium teacher annotator validation of a sample of translated MILU items to estimate translation-induced answer-key error rate (addresses Output Content and Input Content concerns); (5) verification that production inference mode matches the evaluation method used (log-likelihood vs. generative) per [\[Q85\]](#).

ID	Evidence
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's taxonomy of 8 domains and 41 subjects spans the breadth needed for a multi-subject teacher-evaluation deployment, and the format is exclusively single-correct-answer MCQ, which is a direct structural match for the deployment's binary positive/negative marking requirement. However, the taxonomy is derived from competitive exam content (UPSC/SSC/state PSC) rather than school-board syllabi, and the dataset sample shows a heavy skew toward Engineering & Tech (23% of Hindi validation items) with under-representation of Civics/Political Science and Panchayati Raj content that is central to UP/MP/Rajasthan/Bihar state-board curricula. Per-subject statistics for Hindi specifically are not publicly available, making it impossible to verify whether the categories adequately cover state-board curricular framing.

Strengths

- Exclusive single-answer MCQ format directly matches the deployment's binary scoring requirement [Q22, Q33, DATASET-D1]
- All 8 declared domains appear in the Hindi validation sample, providing nominal multi-domain breadth [DATASET-D1, D2, D4, D9, D20]
- Indic LLM sub-taxonomy is treated separately, indicating awareness of system heterogeneity [Q60]

ID	Evidence
Q22	"MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge." (p.3)
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q60	"We evaluate around 16 Indic language LLMs on MILU. These models are primarily built by adapting English LLMs, such as LLAMA-2-7B, by first continually pretraining on small amount Indic language data, followed by optionally instruction finetuning them." (p.6)
D1	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गतिबिधुत अधिक होती है — Engineering question about DC series motor — translated from English
D2	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D4	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र वद्रोह' शब्द संविधान में कब जोड़ा गया? 44वे संविधान संशोधन अधिनियम द्वारा — Constitutional law MCQ — Law & Governance domain
D9	नमिनलखिति में से कौन प्रत्यक्ष कर का उदाहरण है? संपत्तिकर — Economics/taxation MCQ — pan-India competitive exam register
D20	मई 2022 में इराक के सुलेमानिया में आयोजित तीरंदाजी एशिया कप 2022 स्टेज 2 अभियान में भारत ने कतिने स्वर्ण पदक जीते? आठ — Sports GK — India-centric, recent current affairs

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment

IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include Hindi Literature (canonical board authors), History (regional medieval and freedom movement), Civics/Political Science with Panchayati Raj and state administration, Geography of North India, Science/Maths in Hindi, Economics, and General Knowledge. MILU declares all 8 domains and 41 subjects [Q42] which nominally cover this breadth, with subject-level statistics in supplementary Table 9 [Q98].

IO-2: MILU's taxonomy is derived from competitive exam content [Q23, Q24, Q27] rather than school/university board syllabi. State-specific civics (Panchayati Raj, state legislatures) is not visible as a dedicated category in the sample [DATASET-D4 is the only Law & Governance item, and concerns the 44th constitutional amendment, not state-level civics]. The single dedicated Hindi-board literature register (close reading of Tulsidas/Premchand/Kabir) appears absent from the sample [DATASET-D5 — the only Literature item is about Wordsworth].

IO-3: Engineering & Tech dominates the Hindi validation sample (6 of 26 items) including DC motors, transformers, rectifiers, and FORTRAN [DATASET-D1, D13, D14, D23], which are characteristic of polytechnic/engineering entrance content rather than secondary or degree-college teacher domains. This represents construct-irrelevant variance for a general Hindi-medium teacher deployment.

IO-4: Identified gaps: (a) school-literature framing for Hindi canonical authors absent; (b) state-specific civics/Panchayati Raj under-represented; (c) over-representation of engineering content; (d) per-language per-subject distributions not published [NOT FOUND per regional YAML state_wise_exam_source_distribution_for_hindi].

ID	Evidence
Q23	"This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others." (p.3)
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q27	"Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years." (p.3)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q98	"Table 9: Detailed subject level statistics of MILU across different languages." (p.20)
D1	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
D4	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? 44वें संविधान संशोधन अधिनियम द्वारा — Constitutional law MCQ — Law & Governance domain
D5	विलियम वर्ड्सवर्थ _____ के कवि हैं। इंग्लैंड — Literature question about English poet Wordsworth — not Hindi-board canonical literature
D13	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: पल्सेटिंग डीसी — Electrical engineering MCQ — translated from English
D14	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D23	फॉरट्रान 77 के फकिस्ड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉर्मेट लेबल — Computer science MCQ — translated, technical vocabulary
WEB-9	HuggingFace dataset is gated, limiting independent verification of per-subject Hindi distributions (https://huggingface.co/datasets/AI4Bharat/MILU)
WEB-12	GitHub subject categories include 'Literature and Linguistics' and 'Arts and Culture' but no item-level framing detail published (https://github.com/AI4Bharat/MILU)

Information Gaps

- Per-language per-subject item counts for Hindi not published in accessible form (Table 9 in gated PDF)
- Whether the test split's domain distribution matches the sampled validation distribution
- Whether MILU includes any items framed in school-literature register vs. competitive-exam GK

Requires Expert Verification

- Whether the 41-subject taxonomy maps cleanly onto UP/MP/Rajasthan/Bihar board syllabi at the subject level
 - Whether the missing categories (state civics, Hindi literary close reading) are material to teacher use cases or marginal
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Remediation

Gap 1: Heavy skew toward Engineering & Tech (23% of Hindi sample) and under-representation of Civics/Political Science, Panchayati Raj, and state-board literature framings

Recommendation: Use MILU domain-level results as a coarse signal only; build a deployment-specific subject inventory that mirrors UP/MP/Rajasthan/Bihar board syllabi, and weight or filter MILU subjects to align with actual teacher use cases

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content is the highest-priority dimension per elicitation (HIGH), and dataset analysis reveals a critical concern: all 26 sampled Hindi validation examples are flagged is_translated=True, suggesting the Hindi validation split is dominated by GPT-4O-translated English-source items rather than natively-sourced Hindi competitive-exam questions. While MILU's documentation claims an India-first design [Q11] and includes regionally relevant subjects [Q12, Q14], the actual Hindi content sampled shows: (a) literary content focused on Wordsworth rather than Tulsidas/Premchand/Kabir [DATASET-D5]; (b) translated technical terminology using transliterations (शेल, नाइट्रेट, जंप लेबल) that may not align with state-board Hindi textbook conventions [DATASET-D14, D15, D23]; (c) only one item with explicit North Indian state-specific content (UP minerals) [DATASET-D2]; (d) parallel translated items appearing across all 11 language configs [DATASET-D11, D18, D19, D24]. The paper's dataset-level 25% translation figure [Q47] underestimates the apparent Hindi-validation-split translation rate.

ID	Evidence
Q11	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q14	"We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)
Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D2	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D5	विलियम वर्ड्सवर्थ _____ के कवि हैं। इंग्लैंड — Literature question about English poet Wordsworth — not Hindi-board canonical literature
D11	What was the first capital of the Bahamani Kingdom? ... Gulbarga — Medieval Indian history — source question in English
D14	स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D15	रक्त में नमिनलखिति में से किसकी अधिकता 'ब्लू बेबी सड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को द्रगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check
D23	फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉरमेट लेबल — Computer science MCQ — translated, technical vocabulary
D24	एक भौगोलिक क्षेत्र में नमिनलखिति वशिष्ट वशिषताएँ हैं: ... उत्तरी अमेरिका का अटलांटिक तट — Mediterranean climate geography — translated, no North India content

Strengths

- Documented India-first design intent with stated coverage of local history, arts, festivals, and laws [Q2, Q11, Q12]
- Region-specific exams are explicitly emphasized for capturing local knowledge [Q14, Q28]

- Some India-centric Hindi items present (UP minerals, Mughal history, Amrita Shergil, Radcliffe Line) [DATASET-D2, D3, D8, D25]
- Cross-language answer-key consistency demonstrated for translated parallel items [DATASET-D18, D19]

ID	Evidence
Q2	“With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science.” (p.1)
Q11	“We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.” (p.2)
Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)
Q14	“We focus on region-specific exams to authentically capture local knowledge in the respective language.” (p.2)
Q28	“Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state.” (p.3)
D2	नमिन्लखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D3	मुगल ग्रंथ 'मासुरि-ए-आलमगीरी' के रचयिता कौन हैं? साकी मुसुतईद खान — Mughal history — competitive-exam GK register
D8	नमिन्लखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार हैं, जिन्हें कभी-कभी भारत की फ्रीडा काहलो के नाम से जाना जाता है? अमृता शेरगिल — Indian arts & culture — Amrita Shergil, India-centric
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check
D25	रेडक्लिफ रेखा भारत की सीमाओं को नमिन्लखिति में से किस देश के साथ विभाजित करती है? पाकिस्तान — Partition history — India-centric GK

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: The deployment requires region-specific knowledge: North Indian state geography, Hindi-board canonical literature, Panchayati Raj civics, state freedom-movement figures. The Hindi sample contains some India-centric items [DATASET-D2, D3, D25] but a majority of items are translated [DATASET-D28] and may not reflect native Hindi competitive-exam content.

IC-2: Cultural alignment is partially present (Mughal history, Indian arts, partition geography are reasonable matches [DATASET-D3, D8, D25]), but the Hindi-literature register [DATASET-D5 — Wordsworth as the only literature item] does not match Hindi-medium school-board norms. Canonical Hindi literary figures (Tulsidas, Premchand, Kabir, Mahadevi Varma) are not present in the sampled items.

IC-3: Translated items retain English technical transliterations (शेल, नाइट्रेट, प्लूसेटिंग डीसी, जंप लेबल) [DATASET-D14, D15, D23] that may diverge from Vaigyaanik evam Takniki Shabdavali Aayog or state-board conventions. The Wordsworth literature item [DATASET-D5] introduces non-regional content. Translated items effectively test translated-English knowledge rather than native Hindi-register knowledge.

IC-4: INSUFFICIENT DOCUMENTATION — MILU does not document recruitment of Hindi-medium North Indian annotators specifically [NOT FOUND per regional YAML annotator_demographics]; institutional base is IIT Madras + IBM Research India, which does not guarantee North Indian Hindi-medium representation.

IC-5: Documented content issues: (a) ~100% translation rate observed in Hindi validation sample vs. ~25% claimed dataset-wide [Q47, DATASET-D28]; (b) absence of canonical Hindi-board literary content; (c) under-representation of UP/Bihar/Rajasthan/MP state-specific civics; (d) translation terminology not validated against state-board Hindi norms [NOT FOUND per regional YAML hindi_register_standardisation].

ID	Evidence
Q11	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q14	"We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)
Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
Q84	"Third, the scarcity of questions necessitated translating a portion of the dataset." (p.9)
D2	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D3	मुगल ग्रंथ 'मासरि-ए-आलमगीरी' के रचयिता कौन हैं? साकी मुस्तईद खान — Mughal history — competitive-exam GK register
D5	विलियम वर्ड्सवर्थ _____ के कर्वा हैं। इंग्लैंड — Literature question about English poet Wordsworth — not Hindi-board canonical literature
D8	नमिनलखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार थीं, जिन्हें कभी-कभी भारत की फर्दा काहलो के नाम से जाना जाता है? अमृता शेरगिल — Indian arts & culture — Amrita Shergil, India-centric
D14	स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D15	रक्त में नमिनलखिति में से किसकी अधिकता 'ब्लू बेबी सिंड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D23	फॉरट्रान 77 के फकिस्ड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉर्मेट लेबल — Computer science MCQ — translated, technical vocabulary
D25	रेडक्लफि रेखा भारत की सीमाओं को नमिनलखिति में से किस देश के साथ वभिजति करती है? पाकस्तान — Partition history — India-centric GK
D28	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... is_translated: True — is_translated=True flag on all 26 Hindi validation examples reviewed
WEB-19	Bihar's majority population speaks Bhojpuri (24.86%), Maithili (12.55%), Magahi (10.87%) — Khari Boli register may amplify mismatch for this sub-population (https://en.wikipedia.org/wiki/Bihar)

Information Gaps

- Whether the test split is also predominantly is_translated=True (only validation split sampled)
- Hindi-specific translation proportion in MILU (paper reports only dataset-wide 25%)
- State-wise distribution of Hindi exam sources (UP/MP/Rajasthan/Bihar shares)
- Whether Hindi literature items in the un-sampled portion include canonical board authors

Requires Expert Verification

- Whether GPT-4O Hindi translations preserve subject-specific terminology in line with UP/MP/Rajasthan/Bihar state-board textbooks
- Whether Khari Boli register mismatches with Bhojpuri/Maithili-speaking Bihar teachers materially affect comprehension
- Whether 'शेल', 'नाइट्रेट', 'जंप लेबल' style transliterations are acceptable in state-board Hindi technical texts

Remediation

Gap 1: 100% of sampled Hindi validation items are machine-translated; translated technical terminology uses English transliterations of unvalidated alignment with state-board Hindi norms; canonical Hindi-board literary content absent from sample

Recommendation: Stratify any deployment-relevant evaluation by is_translated flag and report performance separately for natively-sourced vs. translated Hindi items; supplement with a curated set of native Hindi-board MCQs covering Hindi

literature canonical authors and verified state-board terminology

Gap 2: No published state-wise breakdown of Hindi exam sources for UP/MP/Rajasthan/Bihar

Recommendation: Request access to the gated HuggingFace dataset metadata (or supplementary Table 8) and audit the Hindi item provenance by source-exam state; if Hindi-belt state-PSC sourcing is thin, supplement with state-PSC items directly

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is well-aligned with the deployment context. MILU is exclusively text-based MCQ in Devanagari script for Hindi [DATASET-D6, D7], which matches the deployment's Hindi/Devanagari text-only modality. Hindi is high-resource within MILU's 11-language scope [Q5, Q76], questions with images, reading comprehension, and >4 options were excluded for uniformity [Q33], and language verification used INDICLID and Unicode filtering [Q34]. Manual review steps [Q32, Q36] add quality assurance. The minor concerns are: (a) one truncated English item observed [DATASET-D30] indicates residual data quality noise; (b) translation-induced register mismatch in technical vocabulary may affect surface form for translated items [DATASET-D14, D15, D23], though this is more accurately a content concern. No script, modality, or signal-distribution mismatch with the deployment.

ID	Evidence
Q5	"Models also perform better in high-resource languages as compared to low-resource ones." (p.1)
Q32	"Initially, we manually review a large sample of questions to detect and eliminate potential sources of noise." (p.4)
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q34	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q36	"As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors." (p.4)
Q76	"The analysis also shows that models perform significantly better in high-resource languages than low-resource ones, highlighting the need for more robust multilingual strategies." (p.8)
D6	दिए गए वाक्य का सही अपरत्यक्ष रूप चुने। प्रबंधक ने अपने सहायक से कहा कि उसे अगले महीने बोनस मल्लिगा। — Hindi grammar (indirect speech) — formal Khari Boli register
D7	'मानसून' शब्द की उत्पत्ति किस भाषा से हुई है? अरबी भाषा — Hindi-language item about etymology of 'monsoon' — GK-style
D14	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D15	रक्त में नमिलखिति में से किसकी अधिकता 'ब्लू बेबी सिंड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D23	फॉरट्रान 77 के फक्सिड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉरमेट लेबल — Computer science MCQ — translated, technical vocabulary
D30	The owner of the textile shop brought a ... Calculator — Truncated/incomplete question — possible data quality issue

Strengths

- Clean Devanagari script throughout the Hindi sample with formal Khari Boli register [DATASET-D6, D7]
- Image-based, reading-comprehension, and >4-option items explicitly excluded for uniformity [Q33]
- Language verification via INDICLID and Unicode filtering ensures script correctness [Q34]
- Hindi is high-resource within MILU's scope, favouring this deployment [Q5, Q76]

ID	Evidence
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Q5	“Models also perform better in high-resource languages as compared to low-resource ones.” (p.1)
Q33	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q34	“To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.” (p.4)
Q76	“The analysis also shows that models perform significantly better in high-resource languages than low-resource ones, highlighting the need for more robust multilingual strategies.” (p.8)
D6	दिए गए वाक्य का सही अप्रत्यक्ष रूप चुनें। प्रबंधक ने अपने सहायक से कहा कि उसे अगले महीने बोनस मिला। — Hindi grammar (indirect speech) — formal Khari Boli register
D7	‘मानसून’ शब्द की उत्पत्ति किस भाषा से हुई है? अरबी भाषा — Hindi-language item about etymology of ‘monsoon’ — GK-style

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution alignment is strong: both benchmark and deployment are text-only Hindi/Devanagari MCQ [Q22, Q33, DATASET-D6]. No modality, resolution, or sensor mismatch.

IF-2: Regional infrastructure can support text-only Hindi MCQ — Devanagari IME and mobile rendering are mature. Connectivity is the main constraint (Bihar 43%, UP 46% general-population internet penetration [WEB-4]; rural school internet at 18% [WEB-5]) but does not affect input-form fidelity per se.

IF-3: MCQ four-option format with single correct answer matches deployment requirement exactly [Q33, DATASET-D1]. Translation-induced terminology choices [DATASET-D14, D15, D23] are a register concern but are scored under input content rather than form.

IF-4: Minor form-level issue: one truncated English item observed [DATASET-D30], suggesting residual noise despite filtering. No systemic form mismatch.

ID	Evidence
Q22	“MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge.” (p.3)
Q32	“Initially, we manually review a large sample of questions to detect and eliminate potential sources of noise.” (p.4)
Q33	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q36	“As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors.” (p.4)
D1	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
D6	दिए गए वाक्य का सही अप्रत्यक्ष रूप चुनें। प्रबंधक ने अपने सहायक से कहा कि उसे अगले महीने बोनस मिला। — Hindi grammar (indirect speech) — formal Khari Boli register
D14	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with ‘शेल’ (shell) transliteration
D15	रक्त में नमिलखिति में से किसकी अधिकता ‘ब्लू बेबी सिंड्रोम’ रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with ‘नाइट्रेट’ transliteration retained

D23	फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉरमेट लेबल — Computer science MCQ — translated, technical vocabulary
D30	The owner of the textile shop brought a ... Calculator — Truncated/incomplete question — possible data quality issue
WEB-4	Bihar internet penetration ~43%, UP ~46% (IAMAI/Kantar 2024) — relevant to deployment but not benchmark form (https://www.iamai.in/sites/default/files/research/Kantar_%20IAMAI%20report_2024_.pdf)
WEB-5	Rural school internet access ~18% — affects deployment delivery, not benchmark input form (https://www.amulyacharan.com/2024/12/17/the-digital-divide-in-education-bridging-the-urban-rural-gap/)

Information Gaps

- Frequency of truncated/incomplete items across the full dataset (only one observed in sample)

Remediation

Gap 1: Residual data quality noise observed (one truncated English item)

Recommendation: Run a lightweight pre-deployment quality filter on Hindi items (minimum stem length, option completeness checks) to surface any analogous cases before relying on MILU scores for model selection

Output Ontology (OO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology is well-aligned with the deployment. Ground-truth labels are single correct answers per MCQ drawn from exam answer keys [\[Q12\]](#), which exactly matches the deployment's binary correct/incorrect requirement with positive/negative marking. The label schema is binary at inference time, and the elicitation explicitly confirmed that 'evaluation schemas across boards and states are described as broadly similar' [\[A4\]](#), so competitive-exam keyed answers are considered compatible. The 41-subject 8-domain organisation [\[Q42\]](#) supports per-subject reporting useful for diagnostic analysis. Domain-level analysis surfaces a known weakness of LLMs in culturally specific areas [\[Q20, Q71, Q77\]](#), which the deployment user should be aware of. The taxonomy was derived bottom-up from ~20K fine-grained tags clustered and manually merged [\[Q39, Q40, Q41\]](#), which is sound methodology, although whether the resulting categories carry any culturally variable correct-answer norms is not analysed.

ID	Evidence
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q20	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q39	"Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap." (p.4)
Q40	"To organize them, we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters." (p.4)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q71	"We analyze the performance of various base and instruct models across multiple domains and languages. Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
Q77	"Additionally, the domain-specific analysis indicates that models perform better in general fields such as STEM while facing challenges in culturally relevant subjects like Arts, Humanities, and Law, highlighting the lack of this knowledge in the current models and datasets." (p.8)

Strengths

- Single-correct-answer ground truth from exam answer keys directly matches binary scoring requirement [\[Q12\]](#)
- 41-subject 8-domain organisation supports diagnostic per-subject reporting [\[Q42\]](#)
- Cross-language answer-key consistency for parallel items [\[DATASET-D18, D19\]](#) confirms reliable label application
- Elicitation confirms competitive-exam keys are broadly compatible with North Indian board norms [\[A4\]](#)

ID	Evidence
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)

Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूरगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are MCQ option indices (option1–option4) with a single target [Q12, DATASET-D1]. This is regionally relevant — Indian competitive and board MCQs use the same single-correct format with positive/negative marking [WEB-1, WEB-2].

OO-2: No regionally specific output categories appear missing for a binary MCQ deployment. The deployment does not require partial-credit, rubric, or multi-label output [A1, A2].

OO-3: No categories visibly encode non-regional values. The user confirmed evaluation schemas are broadly consistent across North Indian boards and competitive exams [A4], reducing the risk that competitive-exam answer keys diverge from board norms.

OO-4: Stakeholder-driven taxonomy redesign is not required — the deployment uses the same single-answer MCQ taxonomy as the benchmark.

OO-5: No major taxonomy issues identified. Minor caveat: domain-level categorisation may aggregate over heterogeneous content (e.g., Arts & Humanities mixing competitive-GK and literature) [Q20, Q77], which could affect interpretation of domain-level scores.

ID	Evidence
Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)
Q20	“Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM.” (p.2)
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q77	“Additionally, the domain-specific analysis indicates that models perform better in general fields such as STEM while facing challenges in culturally relevant subjects like Arts, Humanities, and Law, highlighting the lack of this knowledge in the current models and datasets.” (p.8)
D1	जब एक डीसी सीरीज मोटर बनि लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
WEB-1	Indian competitive exams use single-correct MCQ with positive/negative marking — taxonomically aligned with MILU output schema (https://negativemarkingcalculator.in/)
WEB-2	UPSC and similar exams confirm single-answer MCQ output convention (https://testbook.com/upsc-civil-services/negative-marking-in-upsc)

Information Gaps

- Whether domain-level aggregation hides subject-level mismatches relevant to the teacher deployment

Requires Expert Verification

- Whether competitive-exam answer keys for items in Hindi Literature and Civics agree with state-board mark-scheme conventions in detail

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (label correctness for individual datapoints) carries moderate validity risk because MILU does not document annotator demographics or inter-annotator agreement. Annotation relied primarily on subject experts at online exam portals [Q26] with manual audit by AI4Bharat volunteers [Q87]; institutional base is IIT Madras + IBM Research India [Q10], which does not guarantee Hindi-medium North Indian educational representation. The regional YAML notes 'no published MILU source provides annotator demographic details or inter-annotator agreement statistics'. The elicitation does mitigate this concern: the user confirmed evaluation schemas are 'broadly consistent across boards and competitive-exam settings' [A4], and dataset analysis shows cross-language answer-key consistency for parallel items [DATASET-D18, D19]. However, with ~all sampled Hindi validation items machine-translated [DATASET-D28], the original answer keys derive from English-source items and were applied to GPT-4O translations without documented Hindi-medium subject-expert audit of the translation-key alignment.

ID	Evidence
Q10	"Nilekani Centre at AI4Bharat, Indian Institute of Technology, Madras, IBM Research, India" (p.1)
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check
D28	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... is_translated: True — is_translated=True flag on all 26 Hindi validation examples reviewed

Strengths

- Subject experts at online exam portals provided initial answer accuracy [Q26]
- AI4Bharat volunteer team performed manual audits [Q87]
- Cross-language answer-key consistency observed for parallel items [DATASET-D18, D19]
- Elicitation confirms low expected divergence between competitive-exam keys and North Indian board norms [A4]

ID	Evidence
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels derive from exam answer keys [Q12, Q26]; the elicitation confirms these are broadly compatible with North Indian teacher norms [A4]. However, no Hindi-medium North Indian stakeholder validation is documented.

OC-2: Disagreement risk is moderate: most items (factual GK, science, history) are unlikely to have culturally variable answers, but translated items where GPT-4O may have introduced subtle translation errors [DATASET-D14, D15, D23] could yield Hindi-translation answer-key errors not caught by audit.

OC-3: INSUFFICIENT DOCUMENTATION — MILU does not provide annotator demographic breakdowns (regional background, Hindi-medium vs. English-medium education, subject expertise) [NOT FOUND per regional YAML annotator_demographics]; no inter-annotator agreement statistics are reported.

OC-4: Re-annotation by Hindi-medium North Indian teacher pool would address the documented gap, particularly for translated items and Arts & Humanities content where register matters most.

OC-5: INSUFFICIENT DOCUMENTATION — aggregation methods for the cluster-to-subject manual merging [Q41] do not report how disagreements among annotators were resolved, leaving minority-perspective handling opaque.

OC-6: Documented label issues: (a) annotator demographics opaque; (b) no inter-annotator agreement; (c) translated items inherit English-source keys without documented Hindi-medium subject-expert validation.

ID	Evidence
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D14	स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D15	रक्त में नमिन्नलखिति में से कसिकी अधकिता 'बलू बेबी सडिरोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D23	फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉरमेट लेबल — Computer science MCQ — translated, technical vocabulary
WEB-10	NAACL 2025 paper does not publish annotator demographic details or inter-annotator agreement (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- Annotator demographics (region, language background, subject expertise)

- Inter-annotator agreement statistics
- Whether translated Hindi items underwent Hindi-medium subject-expert key validation
- Disagreement-resolution protocols in cluster-to-subject manual merging

Requires Expert Verification

- Spot-check a sample of translated Hindi answer keys against North Indian teacher judgment
 - Verify that competitive-exam keys for Hindi literature, civics, and regional history items align with state-board mark schemes
-

Remediation

Gap 1: No annotator demographics or inter-annotator agreement published; translated Hindi items inherit English-source keys without documented Hindi-medium subject-expert audit

Recommendation: Conduct a focused re-annotation audit by Hindi-medium North Indian teachers on a stratified sample (~200 items) of MILU Hindi items, prioritising Arts & Humanities, Law & Governance, and translated science items to estimate label-correctness rates

Output Form (OF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is fully aligned. MILU evaluates models using accuracy as the primary metric [Q3, Q15, Q16, Q75], with log-likelihood scoring for non-API models [Q52–Q55] and structured JSON generative output for API-based models [Q56, Q57]. The output form is a single label/answer index, which directly matches the deployment's binary correct/incorrect MCQ scoring requirement. Per-subject, per-language, per-shot result tables [Q99–Q132] provide granular evidence for Hindi-specific performance. The author-acknowledged caveat that log-likelihood evaluation may yield different results than generation-based or chain-of-thought evaluation [Q85] is relevant if the deployment's runtime inference is generative — the user should verify that production inference matches the evaluation method used to characterise the chosen model.

ID	Evidence
Q3	"We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.1)
Q15	"We evaluate 45 different LLMs - a mix of closed proprietary, open-source, and language-specific models- on MILU." (p.2)
Q16	"Our findings suggest that models struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.2)
Q52	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q53	"We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020)." (p.5)
Q54	"Specifically, the log-likelihood of an answer (a) given the question (x), i.e., $\log P(a)$ " (p.5)
Q55	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q56	"The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities." (p.5)
Q57	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q75	"We evaluate 45 different LLMs and find that the majority of LLMs struggle on MILU, with GPT4o achieving the highest average accuracy." (p.8)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
Q99	"Table 11: Detailed subject-wise evaluation for ABHINAND/TAMIL-LLAMA-7B-INSTRUCT-V0.2 on MILU across different languages." (p.22)
Q100	"The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.22)
Q101	"Table 12: Detailed subject-wise evaluation for AI4BHARAT/AIRAVATA on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.23)
Q102	"Table 13: Detailed subject-wise evaluation for BHABHAAI/GAJENDRA-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.24)
Q103	"Table 14: Detailed subject-wise evaluation for COGNITIVE-LAB/AMBARI-7B-BASE-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.25)
Q104	"Table 15: Detailed subject-wise evaluation for COGNITIVE-LAB/AMBARI-7B-INSTRUCT-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.26)

Q105	"Table 16: Detailed subject-wise evaluation for GENVRADMIN/ARYABHATTA-GEMMAGENZ-VIKAS-MERGED on MILU across different languages." (p.27)
Q106	"Table 17: Detailed subject-wise evaluation for MANISHIITG/OPEN-ADITI-V6-LLAMA3 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.28)
Q107	"Table 18: Detailed subject-wise evaluation for NICKMALHOTRA/PROJECTINDUS on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.29)
Q108	"Table 19: Detailed subject-wise evaluation for SARVAMAI/OPENHATHI-7B-HI-V0.1-BASE on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.30)
Q109	"Table 20: Detailed subject-wise evaluation for TENSOIC/KAN-LLAMA-7B-BASE on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.31)
Q110	"Table 21: Detailed subject-wise evaluation for ABHINAND/MALAYALAM-LLAMA-7B-INSTRUCT-V0.1 on MILU across different languages." (p.32)
Q111	"Table 22: Detailed subject-wise evaluation for ABHINAND/TELUGU-LLAMA-7B-INSTRUCT-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.33)
Q112	"Table 23: Detailed subject-wise evaluation for SMALLSTEP/MI-SAL-7B-INSTRUCT-V0.1 on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.34)
Q113	"Table 24: Detailed subject-wise evaluation for SMALLSTEP/MI-SAL-7B-BASE-V0.1 on MILU across different languages." (p.35)
Q114	"Table 25: Detailed subject-wise evaluation for TELUGU-LLM-LABS/TELUGU-LLAMA2-7B-V0-BASE on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.36)
Q115	"Table 26: Detailed subject-wise evaluation for GPT-4o on MILU across different languages. The results reported are for 0-shot experiments." (p.37)
Q116	"Table 27: Detailed subject-wise evaluation for GPT-4O-MINI on MILU across different languages. The results reported are for 0-shot experiments." (p.38)
Q117	"Table 28: Detailed subject-wise evaluation for GEMINI-1.5-PRO on MILU across different languages. The results reported are for 0-shot experiments." (p.39)
Q118	"Table 29: Detailed subject-wise evaluation for GEMINI-1.5-FLASH on MILU across different languages. The results reported are for 0-shot experiments." (p.40)
Q119	"Table 30: Detailed subject-wise evaluation for KRUTRIM-SPECTRE-V2 on MILU across different languages. The results reported are for 0-shot experiments." (p.41)
Q120	"Table 31: Detailed subject-wise evaluation for SARVAMAI/SARVAM-1 on MILU across different languages. The results reported are for 5-shot experiments." (p.42)
Q121	"Table 32: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.2-1B on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.43)
Q122	"Table 33: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.2-1B-INSTRUCT on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.43)
Q123	"Table 34: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-2B on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.44)
Q124	"Table 35: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-2B-IT on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.44)
Q125	"Table 38: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.2-3B-INSTRUCT on MILU across different languages." (p.46)
Q126	"Table 39: Detailed subject-wise evaluation for NVIDIA/NEMOTRON-4-MINI-HINDI-4B-BASE on MILU across different languages. The results reported are for 5-shot experiments." (p.47)
Q127	"Table 42: Detailed subject-wise evaluation for NEULAB/PANGAEA-7B on MILU across different languages. The results reported are for 5-shot experiments." (p.49)
Q128	"Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.52)
Q129	"Table 48: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-27B on MILU across different languages. The results reported are X / Y / Z where X denote the 0-shot, Y denotes 1-shot and Z denotes 5-shot performance respectively." (p.52)

Q130	"Table 52: 0-shot subject-wise evaluation for META-LLAMA/LLAMA-3.1-70B-INSTRUCT on MILU across different languages." (p.55)
Q131	"Table 53: 1-shot subject-wise evaluation for META-LLAMA/LLAMA-3.1-70B-INSTRUCT on MILU across different languages." (p.56)
Q132	"Table 54: Detailed subject-wise evaluation for META-LLAMA/LLAMA-3.1-405B on MILU across different languages. The results reported are for 0-shot experiments." (p.57)

Strengths

- Accuracy metric directly matches binary MCQ correct/incorrect scoring [Q3, Q16]
- Both log-likelihood [Q52–Q55] and structured JSON generative [Q56, Q57] evaluation modes available
- Granular per-subject per-language reporting at 0/1/5-shot [Q99–Q132] enables Hindi-specific performance lookup
- Authors document the log-likelihood vs. generation caveat explicitly [Q85]

ID	Evidence
Q3	"We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.1)
Q16	"Our findings suggest that models struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.2)
Q52	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q53	"We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020)." (p.5)
Q54	"Specifically, the log-likelihood of an answer (a) given the question (x), i.e., $\log P(a x)$ " (p.5)
Q55	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i x))$ " (p.5)
Q56	"The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities." (p.5)
Q57	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality is a single discrete answer choice, matching deployment requirement exactly [Q3, Q53–Q55]. No mismatch.

OF-2: Text-to-speech is not required for this deployment (text-only Hindi MCQ); no concern.

OF-3: Target population is postgraduate-qualified Hindi-medium teachers — high literacy assumed; no accessibility-driven output-form constraint identified.

OF-4: Documented caveat: log-likelihood evaluation may diverge from generation-based or CoT performance [Q85]. If deployment uses generative inference, model selection should be informed by API-mode evaluation rather than non-API log-likelihood numbers.

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Remediation

Gap 1: Log-likelihood evaluation may diverge from generative inference (acknowledged by authors)

Recommendation: If production deployment uses generative inference, base model selection on the API-mode (generative) MILU results rather than log-likelihood numbers, or run an internal generative re-evaluation on the candidate model